Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The correct order of magnetic moments (spin only values in B.M.) among is

(A) $[Fe(CN)_6]^{4-} > [MnCl_4]^{2-} > [CoCl_4]^{2-}$

(B) $[MnCl_4]^{2-} > [Fe(CN)_6]^{4-} > [CoCl_4]^{2-}$

(C) $[MnCl_4]^{2-} > [CoCl_4]^{2-} > [Fe(CN)_6]^{4-}$

(D) $[Fe(CN)_6]^{4-} > [CoCl_4]^{2-} > [MnCl_4]^{2-}$
(Atomic nos. : $Mn = 25, Fe = 26, Co = 27$)

Q.2

Outof $[TiF_6]^{2-}$, $[CoF_6]^{3-}$, Cu_2Cl_2 and $[NiCl_4]^{2-}$ (Z of $Ti = 22, Co = 27, Cu = 29, Ni = 28$), the colourless species are:

(A) Cu_2Cl_2 and $[NiCl_4]^{2-}$

(B) $[TiF_6]^{2-}$ and Cu_2Cl_2

(C) $[CoF_6]^{3-}$ and $[NiCl_4]^{2-}$

(D) $[TiF_6]^{2-}$ and $[CoF_6]^{3-}$

Q.3

The basic character of the transition metal monoxides follows the order (Atomic Nos., $Ti = 22,\ V = 23, Cr = 24, Fe = 26$)

(A) $TiO > VO > CrO > FeO$

(B) $VO > CrO > TiO > FeO$

(C) $CrO > VO > FeO > TiO$

(D) $TiO > FeO > VO > CrO$

Q.4

Which of the following statement is incorrect?

(A) Across a period from Sc to Cu the densities increase with increasing atomic number.

(B) The melting point of transition elements rise to a maximum from Sc to Cr and then decreases from Fe to Zn.

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(C) Transition elements have high enthalpies of atomization and in $3d$ series increase regularly from Sc to Cu.

(D) On going down a group from $3d$ to 6 d series the stability of higher oxidation state increases with increasing atomic number.

Q.5

On the basis of data given below, $\mathrm{E}^{\ominus}_{\mathrm{Sc}^{3+}/\mathrm{Sc}^{2+}} = -0.37, \mathrm{E}^{\ominus}_{\mathrm{Mn}^{3+}/\mathrm{Mn}^{2+}} + 1.57$
$\mathrm{E}^{\ominus}_{\mathrm{Cr}^{2+}/\mathrm{Cr}} = -0.90, \mathrm{E}^{\Theta}_{\mathrm{Cu}^{2+}/\mathrm{Cu}} = 0.34$
Which of the following statements is incorrect?

(A) Sc^{3+} has good stability due of $[\mathrm{Ar}]3d^0 4s^0$ configuration.

(B) Mn^{3+} is more stable than Mn^{2+}.

(C) Cr^{2+} is reducing in nature.

(D) Copper does not give H_2 on reaction with dil. $\mathrm{H}_2\mathrm{SO}_4$.

Q.6

A metal M and its compound can give the following observable changes in a consequence of reactions

$$\mathrm{M} \xrightarrow[\mathrm{HNO_3}]{\text{dilute}} \left[\begin{array}{c}\text{Colourless}\\ \text{solutions}\end{array}\right] \xrightarrow[\mathrm{NaOH}]{\text{aqueous}} \left[\begin{array}{c}\text{White}\\ \text{precipitate}\end{array}\right]$$

$$\left[\begin{array}{c}\text{White}\\ \text{precipitate}\end{array}\right] \xleftarrow{\mathrm{H_2S}} \left[\begin{array}{c}\text{Colourless}\\ \text{solutions}\end{array}\right] \xleftarrow[\mathrm{NaOH(aq.)}]{\text{excess}}$$

(A) Mg

(B) Pb

(C) Zn

(D) Sn

Q.7

Which of the statements is not true?

(A) On passing H_2 S through acidified $\mathrm{K}_2\mathrm{Cr}_2\mathrm{O}_7$ solution, a milky colour is observed.

(B) $\mathrm{Na}_2\mathrm{Cr}_2\mathrm{O}_7$ is preferred over $\mathrm{K}_2\mathrm{Cr}_2\mathrm{O}_7$ in volumetric analysis.

(C) $\mathrm{K}_2\mathrm{Cr}_2\mathrm{O}_7$ solution in acidic medium is orange.

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(D) $K_2Cr_2O_7$ solution becomes yellow on increasing the pH beyond 7 .

Q.8

The blue complex formed on addition of conc. NH_4OH solution to a Cu^{2+} salt solution has the structure?

(A) $[Cu(NH_4)_4]^{2+}$

(B) $[Cu(NH_3)_2]^{2+}$

(C) $[Cu(NH_3)_4]^{2+}$

(D) $[Cu(NH_4)_2]^{2+}$

Q.9

When $KMnO_4$ is added to oxalic acid, the decolourisation is slow in the beginning but becomes instantaneous after sometime because

(A) Mn^{2+} acts as autocatalyst.

(B) CO_2 is formed as the product.

(C) Reaction is exothermic.

(D) MnO_4^- catalyses the reaction.

Q.10

Number of electrons transfered in each case when $KMnO_4$ acts as an oxidising agent to give MnO_2, Mn^{2+}, $Mn(OH)_3$ and MnO_4^{2-} are respectively

(A) $3, 5, 4$ and 1

(B) $4, 3, 1$ and 5

(C) $1, 3, 4$ and 5

(D) $5, 4, 3$ and 1

Q.11

Chloro compound of Vanadium has only spin magnetic moment of $1.73BM$. This Vanadium chloride has the formula:

(A) VCl_2

(B) VCl_4

(C) VCl_3

(D) VCl_5

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Q.12

Which pair of elements with the given atomic numbers is expected to have similar properties?

(A) 40,72

(B) 20,36

(C) 10,28

(D) 11,12

Q.13

Which of the following compounds has colour but no unpaired electrons?

(A) $KMnO_4$

(B) K_2MnO_4

(C) $MnSO_4$

(D) $MnCl_2$

Q.14

Which one of the following exhibits the large number of oxidation states?

(A) $Ti(22)$

(B) $V(23)$

(C) $Cr(24)$

(D) $Mn(25)$

Q.15

The element with which of the following outer electron configuration may exhibit the largest number of oxidation states in its compounds :

(A) $3d^5 4s^2$

(B) $3d^8 4s^2$

(C) $3d^7 4s^2$

(D) $3d^6 4s^2$

Q.16

Reacton of zinc with cold and very dilute nitric acid yields

(A) $Zn\,(NO_3)_2 + N_2O$

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(B) $\mathrm{Zn\,(NO_3)_2 + NO}$

(C) $\mathrm{ZnNO_{32} + NH_4NO_3}$

(D) $\mathrm{Zn\,(NO_3)_2 + NO_2}$

Q.17

Of the following outer electronic configuration of atoms, the highest oxidation state is achieved by which one of thern?

(A) $(n-1)d^5ns^2$

(B) $(n-1)d^3ns^2$

(C) $(n-1)\mathrm{d}^5ns^1$

(D) $(n-1)\mathrm{d}^3\ \mathrm{ns}^2$

Q.18

In which of the tollowing ions, d-d transition is not possible ?

(A) T^{4+}

(B) Cr^{3+}

(C) Mn^{2+}

(D) Cu^{2+}

Q.19

Whch of the following pairs involves isoelectronic ions

(A) Mn^{3+} and Re^{2+}

(B) Mn^{2+} and Fe^{3+}

(C) Cr^{3+} and Mn^{2+}

(D) Fe^{2+} and Co^{2+}

Q.20

In a reaction the ferrous $\left(\mathrm{Fe}^{+2}\right)$ ion is oxidised to ferric $\left(\mathrm{Fe}^{+3}\right)$ ion. The equivalent weight of the ion in the above reaction is equal to

(A) Half of the atomic weight

(B) $1/5$ of the atomic weight

(C) The atomic veight

(D) Twice the atomic weight

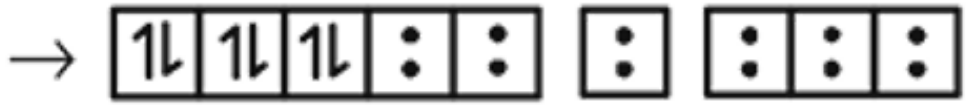
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Q.1 Answer:

$[MnCl_4]^{2-} > [CoCl_4]^{2-} > [Fe(CN)_6]^{4-}$

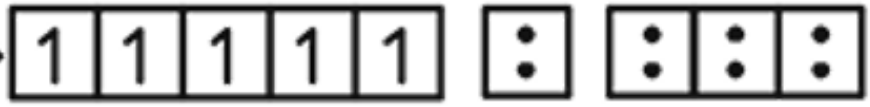
Solution:

$[Fe(CN)_6]^4$



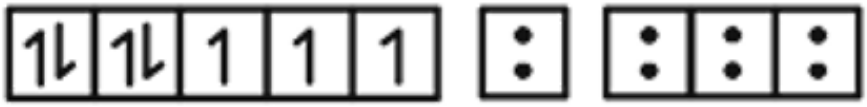
no of unpaired electron $= 0$

$[MnCl_4]^{2-} \rightarrow$



no of unpaired electron $= 5$

$[CoCl_4]^{2-} \rightarrow$



no of unpaired electron $= 3$ The greater the number of unpaired electrons, greater the magnitude of magnetic moment. Hence the correct order will be $[MnCl_4]^{2-} > [CoCl_4]^{2-} > [Fe(CN)_6]^4$

Q.2 Answer:

$[TiF_6]^{2-}$ and Cu_2Cl_2

Solution:

The colour exhibited by transition metal ions is due to the presence of unpaired electrons in d -orbitals which permits the $d-d$ excitation of electrons.

In $[TiF_6]^{2-} - Ti$ is in $+4$ O.S. ; $3d^0$ -colourless

In $[CoF_6]^{3-} - Co$ is in $+3O \cdot S; 3d^5 -$ coloured

In $Cu_2Cl_2 - Cu$ is in $+1$ O.S. ; $3d^{10} -$ colourless

In $[NiCl_4]^{2-} - Ni$ is in $+2O \cdot S; 3d^8 -$ coloured

Q.3 Answer:

$TiO > VO > CrO > FeO$

Solution:

The order of basic character of the transition metal monoxide is $TiO > VO > CrO > FeO$ because basic character of oxides decreases with increase in atomic number.

Q.4 Answer:

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Transition elements have high enthalpies of atomization and in $3d$ series increase regularly from Sc to Cu.

Solution:

(a), (b) and (d) are correct statements.

(c) There is drop in enthalpy of atomization at manganese because of half filled stable electronic configuration leading to less mobilisation of electrons. This results in weaker metallic bonds.

Q.5 Answer:

Mn^{3+} is more stable than Mn^{2+}.

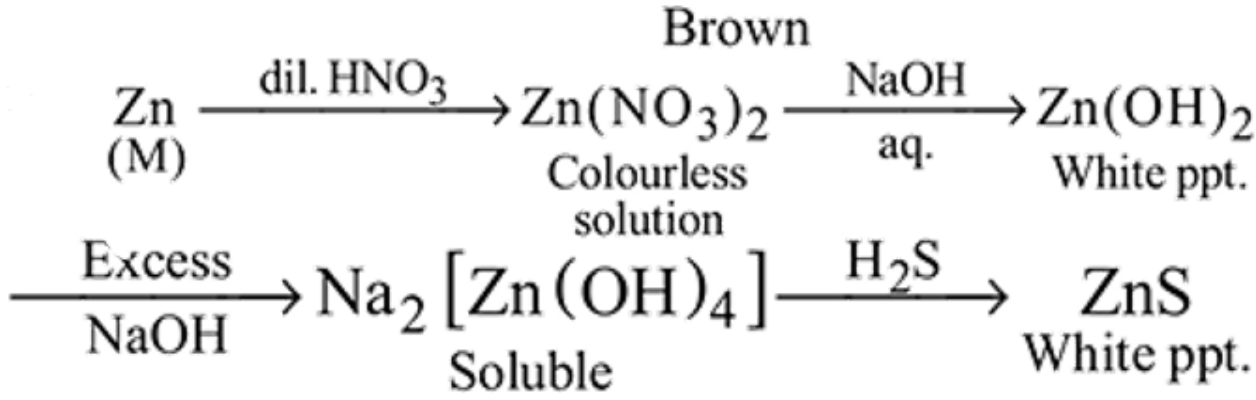
Solution:

$\mathrm{Mn}^{2+}\left(d^5\right)$ **is more stable than** $\mathrm{Mn}^{3+}\left(d^4\right)$**, thus** $\mathrm{E}_{\mathrm{Mn}^{3+}/\mathrm{Mn}^{2+}} = +\mathrm{ve}$

Q.6 Answer:

Zn

Solution:



Q.7 Answer:

$\mathrm{Na_2Cr_2O_7}$ is preferred over $\mathrm{K_2Cr_2O_7}$ in volumetric analysis.

Solution:

$\mathrm{Na_2Cr_2O_7}$ **is hygroscopic.**

Q.8 Answer:

$\left[\mathrm{Cu}\left(\mathrm{NH_3}\right)_4\right]^{2+}$

Solution:

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$$Cu^{2+} + 4NH_4OH \rightarrow [Cu(NH_3)_4]^{2+} + 4H_2O$$

deep blue comp

Q.9 Answer:

Mn^{2+} acts as autocatalyst.

Solution:

Little bit acid is required for reaction $\therefore$ Mn^{2+} acts as autocatalyst.

Q.10 Answer:

$3, 5, 4$ and 1

Solution:

	$KMnO_4$	MnO_2	Mn^{2+}	$Mn(OH)_3$	MnO_4^{2-}
O.S of Mn	+7	+4	+2	+3	+6
electrons needed	0	3	5	4	1

Q.11 Answer:

VCl_4

Solution:

$\mu = \sqrt{n(n+2)}$

$1.73 = \sqrt{n(n+2)}$

On calculating the value of n we find $n = 1$ No. of unpaired electrons $= 1$ hence its configuration will be $V(23) = [Ar]3d^3 4s^2$

$V^{4+} = [Ar]3d^1$

$\therefore$ Its chloride has the formula VCl_4

Q.12 Answer:

40,72

Solution:

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$Zr(40), Hf(72)$ **have similar properties as they belong to same group and have same valence shell electronic configuration.**

Q.13 Answer:

$KMnO_4$

Solution:

Electronic configuration of Mn **is** $[Ar]3d^5 4s^2$**. Being transition metal it has 7 valence electrons and all are involved in bond formation in** MnO_4^-**. Hence it has no unpaired electron.**

Q.14 Answer:

$Mn(25)$

Solution:

Manganese exhibits the large number of oxidation states. The most common oxidation states of Mn **are** $+2, +3, +4, +6$ **and** $+7$**.**

Q.15 Answer:

$3d^5 4s^2$

Solution:

The element with outer electron configuration $3d^5 4s^2$ **is Mn which exhibits oxidation states from** $+2$ **to** $+7$**.**

Q.16 Answer:

$ZnNO_{32} + NH_4NO_3$

Solution:

Very dil HNO_3 $:\longrightarrow$ **Ammonium nitrate is formed**

$4Zn + 10HNO_3 \longrightarrow 4Zn\,(NO_3)_2 + NH_4NO_3 + H_2O$

Q.17 Answer:

$(n-1)d^5 ns^2$

Solution:

For the elements of transition series in which (n-1)d subshell is not more than half-filled, the maximum oxidation state is equal to the sum of ns and (n-1)d electrons.
For the elements where $(n-1)d$ **shell is more than half-filled, the maximum oxidation state is not related to their electronic configuration.**

Q.18 Answer:

T^{4+}

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Solution:

Due to absence of d-electron in $\mathrm{Ti}^{4+}\left([\mathrm{Ar}]^{18} 3\ \mathrm{d}^{0} 4\ \mathrm{s}^{0}\right), \mathrm{d}-\mathrm{d}$ transition is not possible.

Q.19 Answer:

Mn^{2+} and Fe^{3+}

Solution:

Among the given choice Mn^{2+} and Fe^{3+} involve isoelectronic ions.

Q.20 Answer:

The atomic veight

Solution:

The atomic weight :

Equivalent weight $= \frac{\text{Atomic weight}}{\text{No. of e lost or gained}}$

$\mathrm{Fe}^{2+} \longrightarrow \mathrm{Fe}^{3+} + \mathrm{e}^{-}$

$\therefore$ **Equivalent weight = Atomic weight**

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